

Problem Statement

Flooding caused by rising dam water levels poses a serious threat to human life, agriculture, and infrastructure. Traditional methods of monitoring water levels often require manual observation, which can be slow, unreliable, and unsafe during extreme weather conditions.

The objective of this project is to develop a microcontroller-based Dam Water Level Alert System that continuously monitors the water level using an HC-SR04 ultrasonic sensor. The system displays the measured water level on a 16×2 LCD, provides visual alerts using Green, Yellow, and Red LEDs, activates a buzzer during critical conditions, and operates a servo motor to simulate the opening of a flood gate.

The prototype demonstrates how embedded systems can be used to provide real-time monitoring and early warning for water level management in a simple and cost-effective manner.